

3-b Family Facts in Prolog

PROGRAM:

```
% Facts
parent(john, mike).
parent(john, susan).
parent(mary, mike).
parent(mary, susan).

male(john).
male(mike).
female(mary).
female(susan).

% Rules
father(X, Y) :- parent(X, Y), male(X).
mother(X, Y) :- parent(X, Y), female(X).
grandparent(X, Y) :- parent(X, Z), parent(Z, Y).
siblings(X, Y) :- parent(Z, X), parent(Z, Y), X \= Y.

% Interactive query
run_family :-
    write('Enter relation (father/mother/grandparent/siblings): '), read(Rel),
    write('Enter first person: '), read(P1),
    write('Enter second person: '), read(P2),
    (call(Rel, P1, P2) -> writeln('True') ; writeln('False')).
```

OUTPUT:

```
SWI-Prolog console
File Settings Tools Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

8 ?- consult(['c:/Users/kames/OneDrive/Documents/family-facts.pl']).
true.

9 ?- run_family.
Enter relation (father/mother/grandparent/siblings): mother.
Enter first person: |: john.
Enter second person: |: mike.
false
true.

10 ?- run_family.
Enter relation (father/mother/grandparent/siblings): father.
Enter first person: |: john.
Enter second person: |: mike.
True
true.

10 ?- ▲
```